

DevOps Lab

Documentation

Program 7:

Deploy a Web Application to Kubernetes using Minikube

The program demonstrates how to deploy a simple web application (Nginx) on a local Kubernetes cluster using Minikube. The aim is to understand the process of creating a Deployment, exposing it through a Service and accessing the web application running inside Kubernetes pods.

Project Structure:

```
program_7/  
—>nginx-deployment.yaml
```

Procedure:

Create a folder (ex: program_7)

Step 1: Start Minikube

- minikube start

- Check the status of Minikube

- minikube status

```
Nov 25 21:00
1RV24MC062_lubna_tabassum@Inspiron: ~
1RV24MC062_lubna_tabassum@Inspiron:~$ minikube start
minikube v1.37.0 on Ubuntu 24.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Updating the running docker "minikube" container ...
Failing to connect to https://registry.k8s.io/ from both inside the minikube container and host machine
To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: storage-provisioner, default-storageclass

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
  ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC062_lubna_tabassum@Inspiron:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 2: Create nginx-deployment.yaml file with the following code
nano nginx-deployment.yaml

```
Dec 1 20:07
1RV24MC062_lubna_tabassum@Inspiron: ~/DevOps/program_7
GNU nano 7.2 nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Step 3: Apply the nginx-deployment.yaml file

docker apply -f nginx-deployment.yaml

Check status

docker get pods

docker get deployments

```
1RV24MC062_lubna_tabassum@Inspiron: ~/DevOps/program_7
minikube v1.37.0 on Ubuntu 24.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Updating the running docker "minikube" container ...
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: default-storageclass, storage-provisioner

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
  ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_7$ nano nginx-deployment.yaml
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-knwgq   1/1     Running   0           7s
nginx-deployment-6f9664446b-zggtq   0/1     ContainerCreating   0           7s
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_7$ kubectl get deployments
NAME            READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 2/2      2             2           11s
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_7$ kubectl expose deployment nginx-deployment
```

Step 4: Expose the deployment as service

kubect expose deployment nginx-deployment - -type=NodePort - -port=80

Check services

kubectl get services

Acces the web app

minikube service nginx-deployment

```
nginx-deployment 2/2 2
1RV24MC062_lubna_tabassum@Inspiron:~$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_7$ kubectl get services
NAME          TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes    ClusterIP     10.96.0.1     <none>         443/TCP          8m35s
nginx-deployment NodePort      10.110.154.169 <none>         80:32649/TCP     7s
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_7$ kubectl service nginx-deployment
error: unknown command "service" for "kubectl"
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:32649

```
Opening service default/nginx-deployment in default browser...
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_7$ WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1764149210.387743 22028 voice_transcription.cc:58] Registering VoiceTranscriptionCapability
Created TensorFlow Lite XNNPACK delegate for CPU.
Attempting to use a delegate that only supports static-sized tensors with a graph that has dynamic-sized tensors (tensor#-1 is a dynamic-sized tensor).
[22003:22033:1126/145650.908827:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[22003:22033:1126/145650.908827:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registrat
```

Output

